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In this sheet, we will discuss a few (counterintuitive) facts about high-dimensional probability. The reason to study this matter is twofold: First, many machine learning problems are high-dimensional, either in the data space or in the parameter space. Second, the behavior of random variables in high dimensions can be quite different from what we expect in low dimensions. We consider random variables

$$X^{(d)} \sim \mathcal{N}(0, I_d),$$

where $d \in \mathbb{N}$ denotes the dimension of the ambient space.

Task 1 (Refresher on Gaussian random variables)

Compute

1. $\mathbb{E}X^{(d)}$, $\operatorname{Cov} X^{(d)}$,
2. $\mathbb{E}\left\|X^{(d)}\right\|_2^2$, $\operatorname{Var}\left\|X^{(d)}\right\|_2^2$,

where $\|\cdot\|_2$ denotes the Euclidean norm.

Task 2 (Thin shell phenomenon)

Consider the following version of Chebyshev's inequality: Let Y be a real-valued random variable and let $a > 0$. Then

$$\mathbb{P}(|Y - \mathbb{E}Y| \geq a) \leq \frac{\operatorname{Var} Y}{a^2}.$$

We want to use it to show that $X^{(d)}$ does not concentrate around its mean.

1. Show that $|\left\|X^{(d)}\right\|_2^2 - d| \geq \sqrt{d}|\left\|X^{(d)}\right\|_2 - \sqrt{d}|$.
2. Use Chebyshev to conclude that

$$\mathbb{P}\left(\left|\left\|X^{(d)}\right\|_2 - \sqrt{d}\right| \geq \epsilon\right) \leq \frac{2}{\epsilon^2}, \quad \epsilon > 0.$$

Interpret the result!

Task 3 (Visualisation of the thin shell phenomenon)

Consider $d \in \{1, 2\}$. For each case, generate a few hundred (independent) samples of $X^{(d)}$.

1. Make a d-dimensional scatter plot with the samples and indicate a circle of radius $\sqrt{d}$ as well as the mean.
2. Make a one-dimensional histogram of the samples of $\left\|X^{(d)}\right\|_2$ and indicate $\sqrt{d}$. Make this plot also for higher dimensions, e.g. $d \in \{100, 1000, ...\}$.

Bonus: Based on Task 2, add a confidence band to the histogram plots. Is the bound of Task 2 tight?